Infinitesimal Calculus 2, Moed B

2024, Reichman university.

Do not write at the back of the papers!!

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Instructions:

1. Duration: 3 hours.

   Instructions:

   The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits.

   You must explain in full details all of your answers in all parts of the exam, and state all of the conditions for every theorem and/or claim. **Unexplained answers will not receive any credit. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

1. **(25 points)**

1.1. (15 points) Let $a, b \in \mathbb{R}$ such that $a < b$. Let f, g be two functions that are defined on $[a, b]$. Suppose that

i. f, g are continuous on $[a, b]$.

ii. f, g are differentiable on (a, b).

iii. $g'(x) \neq 0$ for every $x \in (a, b)$.

Prove that there exists a $a < c < b$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Solution:

See lectures.

1.2. (10 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Suppose that the series $\sum_{n=1}^{\infty} a_n$ converges absolutely. Prove that the series $\sum_{n=1}^{\infty} \frac{3n-16}{2n+3} a_n$ is convergent.

Solution: It is sufficient to show that the series converges absolutely. We use the comparison test

$$\sum_{n=6}^{\infty} \left| \frac{3n-16}{2n+3} a_n \right| = \sum_{n=6}^{\infty} \frac{3n-16}{2n+3} |a_n| \leq \sum_{n=6}^{\infty} \frac{3n}{2n+3} |a_n| \leq \sum_{n=6}^{\infty} \frac{3n}{2n} |a_n| = \frac{3}{2} \sum_{n=6}^{\infty} |a_n| < \infty$$

It follows that $\sum_{n=6}^{\infty} \left| \frac{3n-16}{2n+3} a_n \right| < \infty$ and therefore $\sum_{n=1}^{\infty} \left| \frac{3n-16}{2n+3} a_n \right| < \infty$.

2. **(25 points)**

2.1. (17 points) Let $(a_n)_{n=1}^{\infty}, (b_k)_{k=1}^{\infty}$ be two sequences of real numbers.

2.1.1. (7 points) Let $(n_k)_{k=1}^{\infty}$ be a strictly increasing sequence of natural numbers. Suppose that $\lim_{n\to\infty} a_n$ exists. Prove that $\lim_{k\to\infty} a_{n_k}$ exists and

$$\lim_{k\to\infty} a_{n_k} = \lim_{n\to\infty} a_n$$

Solution:

See lectures.

2.1.2. (2 points) Define the term: the series $\sum_{k=1}^{\infty} b_k$ is obtained from $\sum_{n=1}^{\infty} a_n$ by applying parentheses.

Solution:

See lectures.

2.1.3. (8 points) Suppose that $\sum_{k=1}^{\infty} b_k$ is obtained from $\sum_{n=1}^{\infty} a_n$ by applying parentheses. Suppose also that the series $\sum_{n=1}^{\infty} a_n$ converges. Prove that $\sum_{k=1}^{\infty} b_k$ converges, and $\sum_{k=1}^{\infty} b_k = \sum_{n=1}^{\infty} a_n$.

Solution: See lectures.

2.2. (8 points) Compute the value of the series $\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+3)}$ or prove that it doesn't converge. Hint: For every $n \in \mathbb{N}$ we have $\frac{1}{(n+1(n+3)} = \frac{1}{2}\left(\frac{1}{n+1} - \frac{1}{n+3}\right)$.

Solution: We note that for every $n \in \mathbb{N}$ we have

$$\frac{1}{(n+1)(n+3)} = \frac{1}{2}\left(\frac{1}{n+1} - \frac{1}{n+3}\right)$$

and therefore

$$\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+3)} = \sum_{n=1}^{\infty} \frac{1}{2}\left(\frac{1}{n+1} - \frac{1}{n+3}\right) = \frac{1}{2} \lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n+1} - \frac{1}{n+3}\right) = \frac{1}{2} \lim_{N\to\infty} \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{N+2} - \frac{1}{N+3}\right) = \frac{5}{12}$$

3. **(25 points)**

3.1. (12 points) Compute the following limit or prove that it doesn't exist in the extended sense

$$\lim_{x\to 0}\frac{\cot(x)-\frac{1}{x}}{x}$$

Note: $\cot(x)=\frac{\cos(x)}{\sin(x)}$.

Solution: We have

$$\begin{aligned}\lim_{x\to 0}\frac{\cot(x)-\frac{1}{x}}{x}&=\lim_{x\to 0}\frac{\frac{\cos(x)}{\sin(x)}-\frac{1}{x}}{x}=\lim_{x\to 0}\frac{x\cos(x)-\sin(x)}{x^2\sin(x)}\overset{\frac{0}{0}}{=}\lim_{x\to 0}\frac{\cos(x)-x\sin(x)-\cos(x)}{2x\sin(x)+x^2\cos(x)}\\&=\lim_{x\to 0}\frac{-\sin(x)}{2\sin(x)+x\cos(x)}\overset{\frac{0}{0}}{=}\lim_{x\to 0}\frac{-\cos(x)}{2\cos(x)+\cos(x)-x\sin(x)}\overset{\text{AOL}}{=}-\frac{1}{3}\end{aligned}$$

3.2. (13 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Consider the sequence defined by

$$b_n=\begin{cases}a_k & n=2k-1\\ -a_k & n=2k\end{cases}\qquad \forall n\in\mathbb{N}$$

Prove that the series $\sum_{n=1}^{\infty}b_n$ is convergent if and only if $\lim_{n\to\infty}b_n=0$.

Solution: First, if the series $\sum_{n=1}^{\infty}b_n$ is convergent then $\lim_{n\to\infty}b_n=0$ by the necessary criterion. Now, suppose that $\lim_{n\to\infty}b_n=0$. Then $\lim_{k\to\infty}a_k=0$. Consider the sequence defined by

$$S_N=\sum_{n=1}^{N}b_n=b_1+b_2+\dots b_N,\qquad \forall N\in\mathbb{N}$$

Note that

$$S_{2N}=a_1-a_1+a_2-a_2+\cdots+a_N-a_N=0,\qquad \forall N\in\mathbb{N}$$

and therefore $\lim_{N\to\infty}S_{2N}=0$. Note also that

$$S_{2N}=S_{2N-1}+b_{2N}=S_{2N-1}-a_N\Rightarrow S_{2N-1}=S_{2N}+a_N,\qquad \forall N\in\mathbb{N}$$

and therefore $\lim_{N\to\infty}S_{2N-1}=\lim_{N\to\infty}[S_{2N}+a_N]=0+0=0$. It follows that $\lim_{N\to\infty}S_N=0$ and therefore

$$\sum_{n=1}^{\infty}b_n=\lim_{N\to\infty}S_N=0$$

4. **(25 points)**

4.1. (15 points) Let $\alpha\geq 0$ and consider the sequence that is defined by

$$\begin{cases}a_{n+1}=\frac{1}{2}\left(a_n^2+1\right) & n\in\mathbb{N}\\ a_1=\alpha\end{cases}$$

4.1.1. (5 points) Prove that the sequence $(a_n)_{n=1}^{\infty}$ is increasing and conclude that the sequence $(a_n)_{n=1}^{\infty}$ converges in the extended sense.

Solution:

Let $n \in \mathbb{N}$. Then $a_n^2 - 2a_n + 1 = (a_n - 1)^2 \geq 0$. Thus, $a_n^2 + 1 \geq 2a_n$. It follows that $2a_{n+1} = a_n^2 + 1 \geq 2a_n \Rightarrow a_{n+1} \geq a_n$. As $(a_n)_{n=1}^{\infty}$ is increasing then it converges in the extended sense and $\lim_{n\to\infty} a_n = \sup\{a_n : n \in \mathbb{N}\}$.

4.1.2. (10 points) For every $\alpha \geq 0$, compute the value of $\lim_{n\to\infty} a_n$.

Solution:

We first show that if $\alpha > 1$ then $\lim_{n\to\infty} a_n = \infty$. : ATC that the sequence $(a_n)_{n=1}^{\infty}$ is convergent. Then there exists an $L \in \mathbb{R}$ such that $L = \lim_{n\to\infty} a_n$. Thus,

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \frac{1}{2}\left(a_n^2 + 1\right) \stackrel{\text{AOL}}{=} \frac{1}{2}\left(L^2 + 1\right)$$

Thus, $L^2 - 2L + 1 = 0 \Leftrightarrow L = 1$. Contradiction, as $L = \sup\{a_n : n \in \mathbb{N}\} \geq a_1 = \alpha > 1$.

Suppose now that $\alpha \leq 1$. We prove that it is bounded from above by 1, using induction. First, $a_1 = \alpha \leq 1$. Let $n \in \mathbb{N}$ and suppose that $a_n \leq 1$. Then

$$a_{n+1} = \frac{1}{2}\left(a_n^2 + 1\right) \leq \frac{1}{2}(1 + 1) = 1$$

As $(a_n)_{n=1}^{\infty}$ is increasing and bounded from above, then it is convergent. It follows that there exists an $L \in \mathbb{R}$ such that $\lim_{n\to\infty} a_n = L$, and we have seen that $L^2 - 2L + 1 = 0$. Thus, $L = 1$. We conclude that

$$\lim_{n\to\infty} a_n = \begin{cases} \infty & \alpha > 1 \\ 1 & \alpha \leq 1 \end{cases}$$

4.2. (10 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Consider the sequence defined by $S_N = \sum_{n=1}^{N} a_n$ for every $N \in \mathbb{N}$ and consider the sequence defined by $b_n = S_n^2 - 2S_{n-1} + 3$ for every $2 \leq n \in \mathbb{N}$. Suppose that the series $\sum_{n=2}^{\infty} b_n$ is convergent. Prove that the series $\sum_{n=1}^{\infty} a_n$ is divergent. Hint: the equation $x^2 - 2x + 3 = 0$ has no real solution.

Solution: ATC that the series $\sum_{n=1}^{\infty} a_n$ is convergent. Then $\exists s \in \mathbb{R}$ such that $\lim_{n\to\infty} S_n = s$. As the series $\sum_{n=1}^{\infty} b_n$ is convergent, then by the necessary criterion $\lim_{n\to\infty} b_n = 0$. Thus, by AOL,

$$0 = \lim_{n\to\infty} b_n = \lim_{n\to\infty}\left(S_n^2 - 2S_{n-1} + 3\right) = s^2 - 2s + 3$$

Contradiction, as the equation $s^2 - 2s + 3 = 0$ has no real solutions.

5. **(25 points)**

5.1. (15 points) Consider the power series $\sum_{n=0}^{\infty} \frac{x^n}{n+1}$.

5.1.1. (5 points) Find the domain of convergence of the power series.

Solution: Note that for $x = -1$ we obtain a conditionally convergent series as

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \text{Leibniz}$$

and

$$\sum_{n=0}^{\infty} \frac{|(-1)^n|}{n+1} = \sum_{n=0}^{\infty} \frac{1}{n+1} = \sum_{n=1}^{\infty} \frac{1}{n} \stackrel{\text{Harmonic}}{=} \infty$$

Thus, the point $x = -1$ is an endpoint of the domain of convergence, and consequently $R = 1$. Note that as the series diverges for $x = 1$ (similarly) then the domain of convergence is $[-1, 1)$.

5.1.2. (10 points) Compute the value of the series $\sum_{n=0}^{\infty} \frac{1}{3^n(n+1)}$.

Solution:

Consider the function defined by $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n+1}$ for every $-1 < x < 1$. We want to compute $f\left(\frac{1}{3}\right)$. By the previous item, f is well defined. It follows that

$$xf(x) = x\sum_{n=0}^{\infty} \frac{x^n}{n+1} = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}, \qquad \forall -1 < x < 1$$

Using term-by-term differentiation, we obtain

$$[xf(x)]' = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \qquad \forall -1 < x < 1$$

It follows that there exists a $C \in \mathbb{R}$ such that

$$xf(x) = -\ln|1-x| + C \qquad \forall -1 < x < 1$$

Plugging $x = 0$ gives $C = 0$, and therefore for every $-1 < x < 1$ we have $xf(x) = -\ln|1-x|$. Thus, $\frac{1}{3}f\left(\frac{1}{3}\right) = -\ln\left(\frac{2}{3}\right)$. and therefore $f\left(\frac{1}{3}\right) = -3\ln\left(\frac{2}{3}\right)$.

5.2. (10 points) Prove that for every $x > -1$ we have

$$(1+x)\ln(1+x) \geq x - \frac{x^2}{2}$$

Solution: Consider the function defined by $f(x) = (1+x)\ln(1+x) - x + \frac{x^2}{2}$ for every $x > -1$. The function f is differentiable by AOD and

$$f'(x) = \ln(1+x) + 1 - 1 + x = \ln(1+x) + x, \qquad \forall x > -1$$

It follows that $f'(x) \leq 0$ for every $-1 < x \leq 0$ and that $f'(x) \geq 0$ for every $x \geq 0$, and therefore f is decreasing on $(-1, 0]$ and increasing on $[0, \infty)$. It follows that $x = 0$ is a minimizer of f, implying that $f(x) \geq f(0) = 0$ for every $x > -1$. Thus

$$f(x) = (1+x)\ln(1+x) - x + \frac{x^2}{2} \geq 0 \qquad \forall x > -1$$

and therefore

$$(1+x)\ln(1+x) \geq x - \frac{x^2}{2} \qquad \forall x > -1$$